IEEE 14-Bus GFL↔GFM Switching with Project AGSI++

Full-state implementation and present evidence

4 August 2026

Abstract

The study uses the configured IEEE 14-bus case, device parameters, and event chronology; switching remains the project AGSI++. The study profile uses a six-state operational EMF6 SG and 12-state GFL/GFM switching supersets. A compressed ideal-SG reclose gate converges and returns all four IBRs to GFL. The 160-s chronology remains fail-closed at 36.040 s because the paper does not publish the DC-source, post-trip active-power dispatch, or multi-GFM coordination laws needed to reproduce its islanded trajectory. No values or plots are altered to force recovery.

1 Case, index, and switching

The 100-MVA, 60-Hz IEEE 14-bus case has one SG at bus 1 and IBRs at buses 2, 3, 6, and 8. The configured apparent-power dispatch is mapped to P/Q with a documented common angle (`PROJECT_DERIVED`); all device values are read from the source table. The project index is

$$J_V = \frac{|V - V_{ref}|}{0.10}, \quad J_f = \frac{|f - 60|}{0.50}, \quad J_R = \frac{|\mathrm{d}f/\mathrm{d}t|}{1.0}, \quad J_P = \frac{|P_{ref} - P|}{0.20},$$

$$J_{SCR} = \max(0, 3/SCR - 1), \quad J_{lock} = \frac{|v_q|}{0.10}, \quad J_{GRA} = 1 - GRA, \qquad \mathrm{AGSI}^{++} = \frac{1}{7}\sum J_k.$$

Each IBR has its own timer. It commits GFL→GFM above $\Gamma_{on} = 0.65$ for $T_{d,on} = 0.10$ s and GFM→GFL below $\Gamma_{off} = 0.35$ with $GRA = 1$ for $T_{d,off} = 1.00$ s. In the full run, IBR1–2 switch at 20.130 s and IBR3–4 at 20.150 s; the action is local, not broadcast. Coordinated SG-reference handback is a separate declared override used only by the reclose gate.

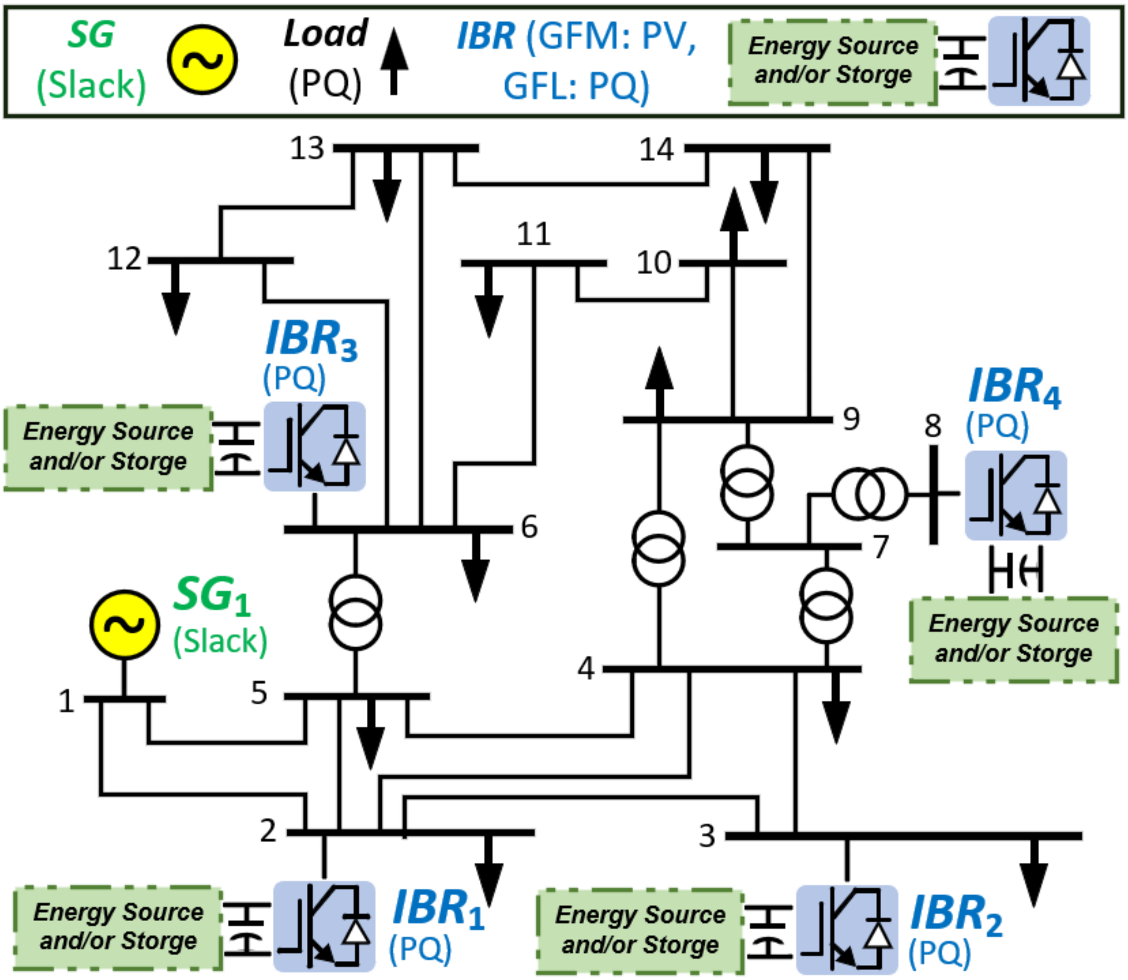

Figure 1: Network and resource locations used by the simulation.

2 Dynamic model

The SG state is $x_{SG} = [\delta, \Delta\omega, E'_q, E'_d, E''_q, E''_d]^T$ with the operational Kundur/EMF6 stator and flux equations. T_m and E_{fd} are equilibrium controls held constant; governor, AVR, synchronizer, and breaker transients are not claimed.

The GFL superset contains filter current, DC-link, PLL, P/Q PI, current PI, and command-delay states. GFM contains filter current, DC-link, virtual angle/speed/EMF, voltage PI, current PI, and delay states. Source values include $L_f = 0.15$, $R_f = 0.015$, $C_{dc} = 0.10$, $I_{max} = 1.2$, $T_d = 0.02$, $k_{p,Idq} = 0.30$, and $k_{i,Idq} = 4.00$. The paper does not publish the energy-source I_{dc} law, so the implementation closes that port with ideal instantaneous power balance; this preserves V_{dc} but cannot establish long-duration energy headroom.

3 Power-flow operating point

Table 1: Bus power-flow results (pu)

Bus	Type	$\lvert V\rvert$	θ (deg)	P_g	Q_g	P_L	Q_L
1	REF	1.06000	+0.0000	+1.31895	-0.26951	0.00000	0.00000
2	PQ	1.05456	-2.9058	+0.31859	+0.17419	0.21700	0.12700
3	PQ	1.03301	-7.6873	+0.28972	+0.15841	0.94200	0.19000
4	PQ	1.04895	-5.9618	+0.00000	+0.00000	0.47800	-0.03900
5	PQ	1.05072	-4.9102	+0.00000	+0.00000	0.07600	0.01600
6	PQ	1.12705	-5.9341	+0.44625	+0.24399	0.11200	0.07500
7	PQ	1.09514	-6.1353	+0.00000	+0.00000	0.00000	0.00000
8	PQ	1.11749	-3.9176	+0.26884	+0.14699	0.00000	0.00000
9	PQ	1.09398	-7.6393	+0.00000	+0.00000	0.29500	0.16600
10	PQ	1.09237	-7.5903	+0.00000	+0.00000	0.09000	0.05800
11	PQ	1.10580	-6.8756	+0.00000	+0.00000	0.03500	0.01800
12	PQ	1.11174	-6.7892	+0.00000	+0.00000	0.06100	0.01600
13	PQ	1.10530	-6.9206	+0.00000	+0.00000	0.13500	0.05800
14	PQ	1.08161	-8.2396	+0.00000	+0.00000	0.14900	0.05000

Table 2: From-end line flows and losses (pu)

Line	From	To	P_{from}	Q_{from}	P_{loss}	Q_{loss}
1	1	2	+0.900928	-0.203057	+0.014518	-0.014696
2	1	5	+0.418019	-0.066451	+0.008475	-0.019813
3	2	3	+0.464318	-0.000621	+0.009133	-0.009246
4	2	4	+0.314334	-0.079982	+0.005358	-0.021354
5	2	5	+0.209348	-0.060569	+0.002332	-0.031219
6	3	4	-0.197092	-0.022968	+0.002456	-0.007604
7	4	5	-0.447907	+0.102332	+0.002561	+0.008079
8	4	7	+0.017005	-0.115901	+0.000000	+0.002495
9	4	9	+0.062330	-0.021423	+0.000000	+0.002062
10	5	6	+0.090091	+0.002266	-0.000000	+0.001610
11	6	11	+0.130988	+0.058687	+0.001540	+0.003226
12	6	12	+0.085939	+0.026725	+0.000784	+0.001631
13	6	13	+0.207419	+0.084233	+0.002610	+0.005140
14	7	8	-0.268840	-0.133746	+0.000000	+0.013242
15	7	9	+0.285846	+0.015351	+0.000000	+0.007516
16	9	10	-0.003741	+0.022196	+0.000013	+0.000036
17	9	14	+0.056917	+0.023544	+0.000403	+0.000857
18	10	11	-0.093754	-0.035839	+0.000693	+0.001622
19	12	13	+0.024155	+0.009094	+0.000119	+0.000108
20	13	14	+0.093845	+0.030080	+0.001359	+0.002767

4 Mode and recovery evidence

Time	Event/index/timer	Committed mode
20.000–20.130 s	SG trip; local on-timers run	IBR1–2 become GFM
20.150 s	remaining local timers mature	IBR3–4 become GFM
36.040 s	full chronology exits the validity domain	fail-closed before later events
1.000–1.140 s (gate)	compressed SG trip and local timers	all GFL→GFM
3.000 s (gate)	ideal SG reference and coordinated handback	all GFM→GFL
4.000 s (gate)	Newton converged; min $\lvert V \rvert$ = 1.033013 pu	all remain GFL

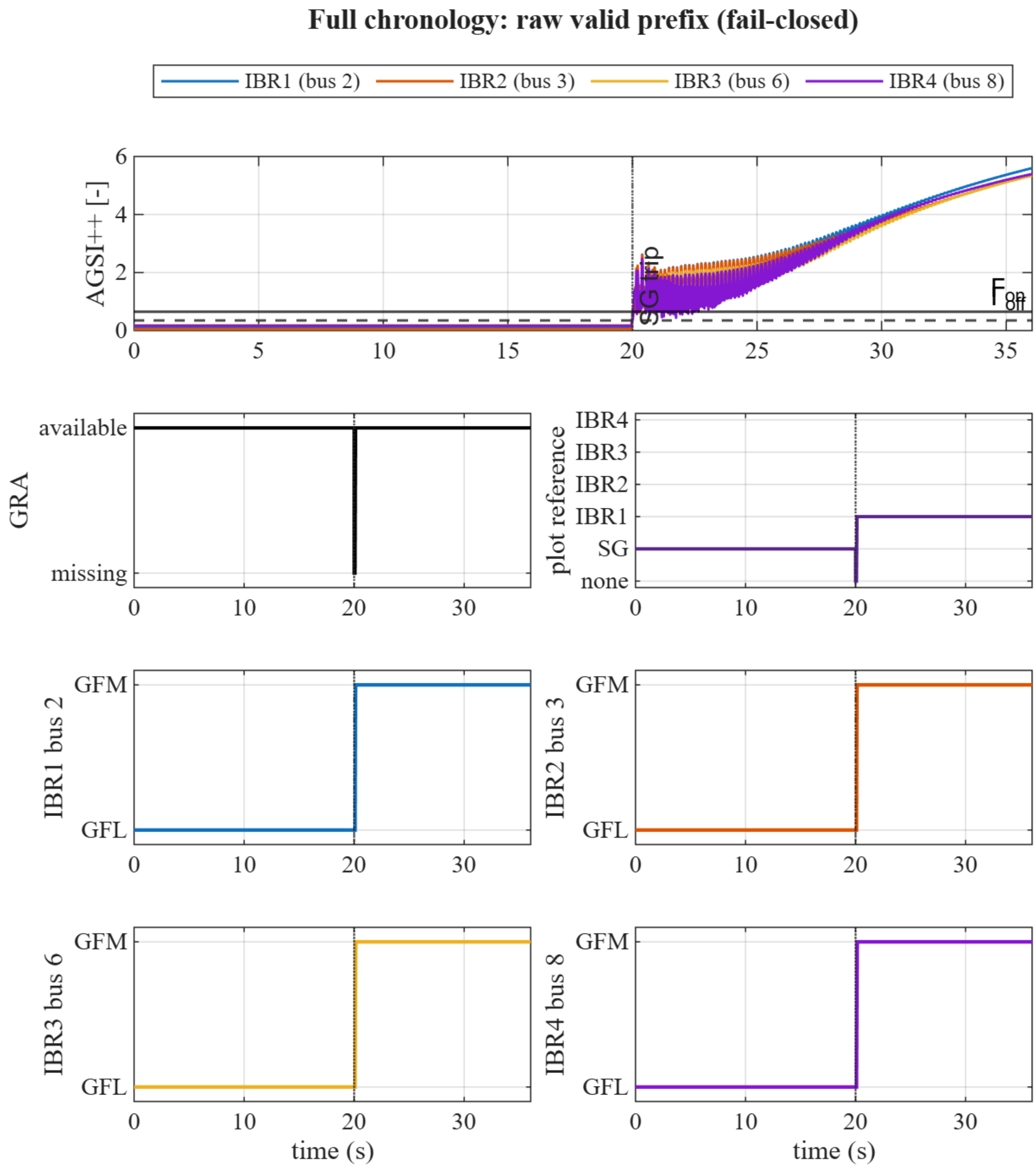

Figure 2: Full-run AGSI++, GRA, reference, and separate 0=GFL/1=GFM timelines.

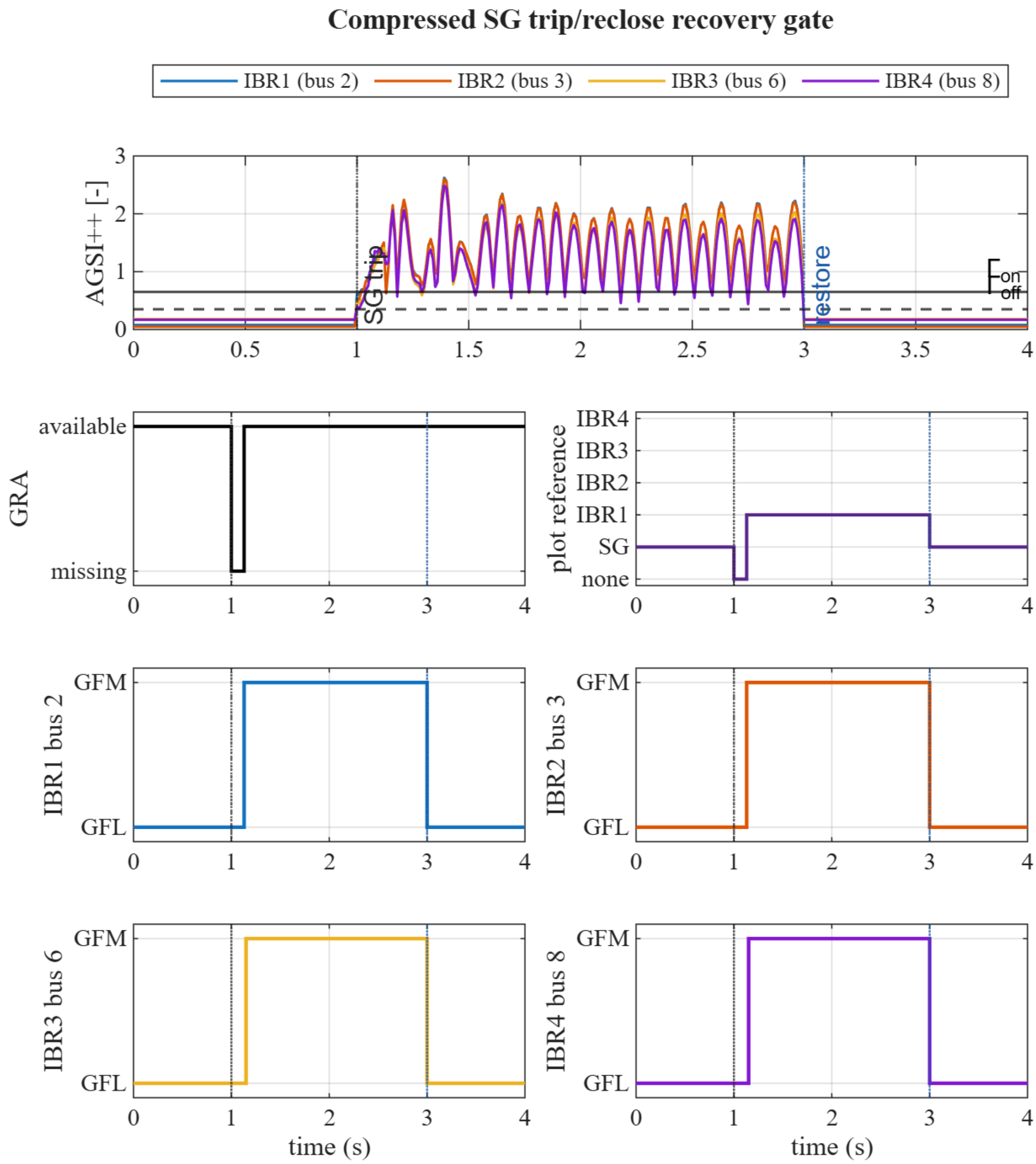

Figure 3: Compressed recovery gate: local switch-up and coordinated SG handback.

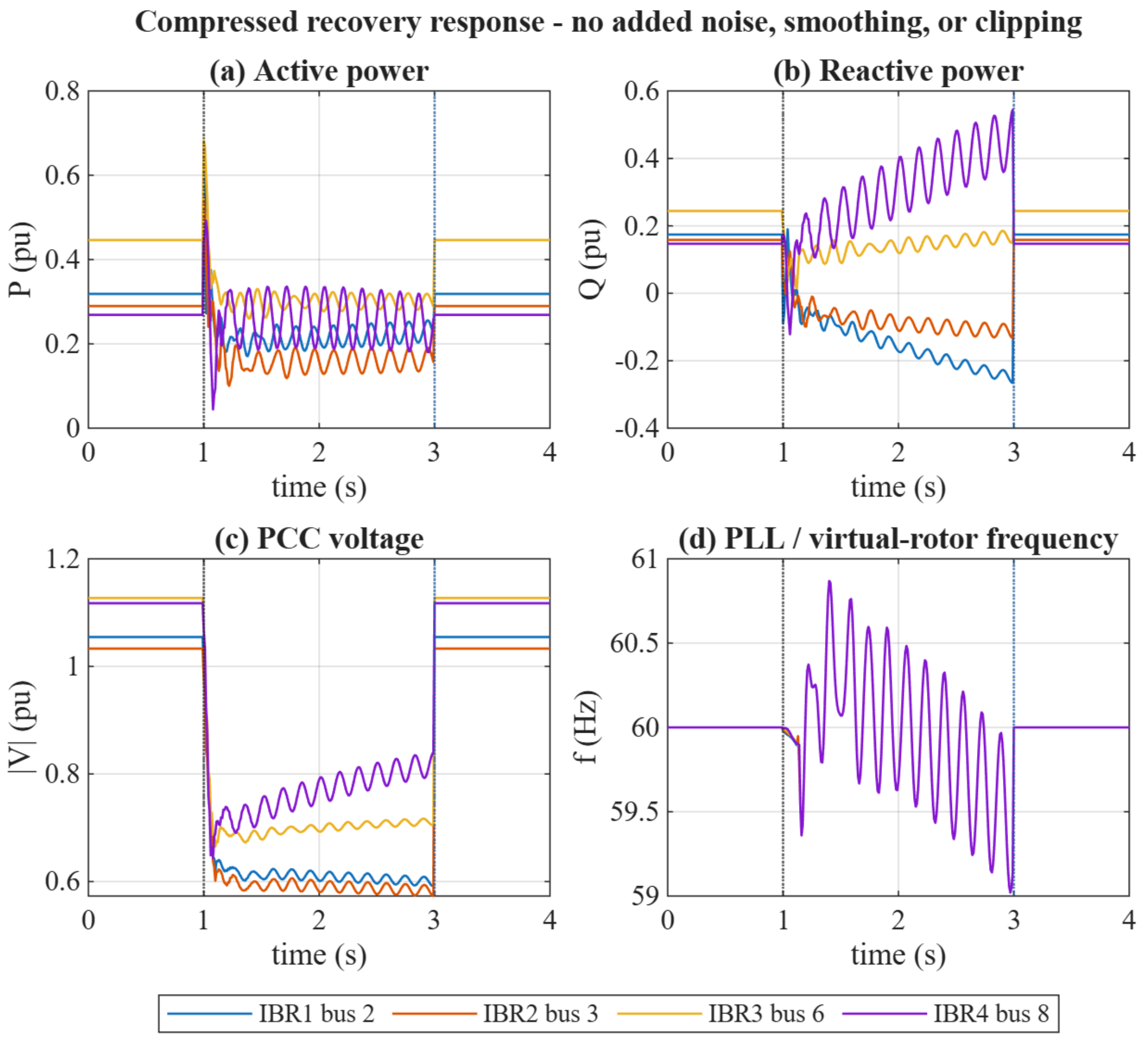

Figure 4: Raw recovery-gate $P, Q, |V|, f$ traces; no noise, smoothing, or clipping.

Existing tests separately cover no-event, light/severe faults, load steps, SG trip, and SG reclose. The long full-state run currently reaches only the SG-trip portion; it does not validate the scheduled 50-s load step, 85-s fault, 110-s line trip, or 145-s restoration. The remaining gap is a missing dispatch/energy/reference-coordination contract, not merely SG state order.

5 Reproduction

```
pf_init_paths;
addpath(fullfile('scripts','reporting'));
generate_ieee14_switch_report_figures;
```

The generator and all solvers are project-owned MATLAB. SSSA and fault-map sections are omitted to keep the report focused on IBR switching.